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Deterministic Chaos

An Introduction

Second revised edition

QC 174.84.S381

VCH

F Shannon's Measure of Information

References: see Chapter 5

This short heuristic introduction into Shannon's measure of information-should enable the reader to understand Chapters 2 and 5. For a more detailed treatment we recommend the book by Shannon and Weaver (1949).

Information Capacity of a Store

Fig. 157a) shows a system with two possible states. If the position of the points is unknown, a priori, and we learn that it is in the left box, say, we gain by definition information amounting to one bit. If we obtain this information, we save one question (with possible answer yes or no which we would have needed to locate the point). Thus, the maximum information content of a system with two states is one bit.

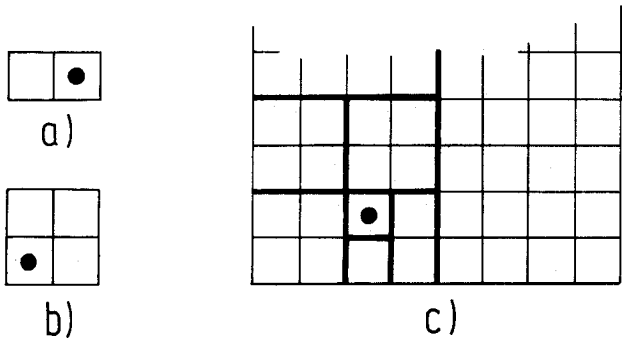


Fig. 157: Information capacity of a store: a) a box with two states. b) It takes two questions (and their answers) to locate a point in a system with four states: right or left? up or down? c) In order to locate a point on a checkerboard with $64 = 2^6$ states, one needs six questions.

For a box with four possible states, one needs two questions to locate the point, i. e. its maximum information content I is

$$I = 2 \text{ (bits)} \tag{F.1}$$

(we will drop the unit "bit" in the following).
This can be written as the logarithm to the base two (1d) of the number of possible states:

$$I = 1\mathrm{d}\, 4\,. \tag{F.2}$$

According to Fig. 157 c, this logarithmic relation between the maximum information content I and the number of states N,

$$I = 1\mathrm{d}\, N \tag{F.3}$$

is true in general.

Information Gain

Let us now calculate the average gain of information if one learns the outcome of statistical events. Suppose we toss a coin such that heads or tails occur with equal probabilities

$$p_1 = p_2 = \frac{1}{2} . \tag{F.4}$$

The information I acquired by learning that the outcome of this experiment is heads, say, is

$$I = 1 \tag{F.5}$$

because there are two equally probable states as in Fig. 157a. This result can be expressed via the $\{p_i\}$ as

$$\mathrm{I} = -\left(\frac{1}{2}\,\mathrm{ld}\,\frac{1}{2} + \frac{1}{2}\,\mathrm{ld}\,\frac{1}{2}\right) \tag{F.6}$$

or

$$I = -\sum p_i\, \mathrm{ld}\, p_i . \tag{F.7}$$

Eq. (F.7) can be generalized to situations where the p_i's are different:

$$p_1 \neq p_2 = 1 - p_1 . \tag{F.8}$$

It then gives the average gain of information if we toss a deformed coin many times.

Let $p_1 = r/q$, where r and q are mutually prime integers, and let us choose the number m of events such that mr/q is again an integer. The total number of distinct states which occur if one tosses a (deformed) coin m times is

$$N = \frac{m!}{(p_1 m)!\,(p_2 m)!} \tag{F.9}$$

where we eliminated, by division, the permutations that correspond to a rearrangement of equal events. The sequences *hht* and *hht* with h = head and t = tail, where the *h's* have been interchanged correspond to the same state. In the limit $m \to \infty$ we can use Stirling's formula, and, for the average information, gain eq. (F.3) yields

$$I = \frac{1}{m} \operatorname{ld} N = \frac{1}{m} \operatorname{ld} \left[\left(\frac{m}{e} \right)^m \left(\frac{e}{p_1 m} \right)^{p_1 m} \left(\frac{e}{p_2 m} \right)^{p_2 m} \right] =$$

$$= - (p_1 \operatorname{ld} p_1 + p_2 \operatorname{ld} p_2) \, . \tag{F.10}$$

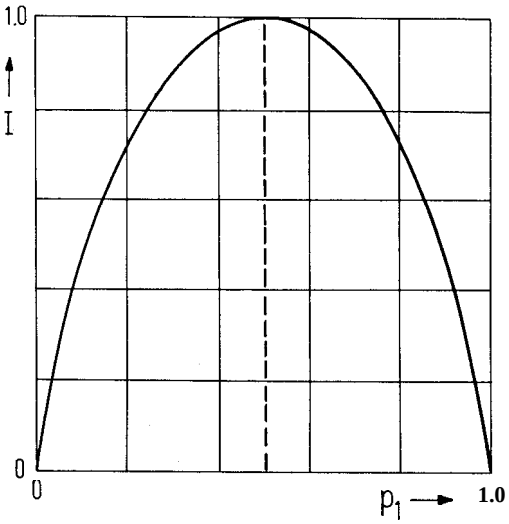


Fig. 158: $I(p)$ for an experiment with two possible outcomes. If $p_1 = 0$, we are sure that the outcome will be event 2, and we gain no information. The maximum information is acquired for $p_1 = p_2 = 1/2$, where the uncertainty of the outcome has its maximum and one learns most from the experiment.

This confirms eq. (F.7) and can be generalized to Shannon's result: If we, a priori, know only that $1 \ldots n$ events (or states of a system) occur with probabilities $\{p_i\}$ (such that $\sum_{i=1}^{n} p_i = 1$) and we learn by a measurement that a certain event j has taken place (or the system occupies actually a certain state), then (if we repeat this measurement many times) we gain the average information

$$I = - \sum_{i=1}^{n} p_i \operatorname{ld} p_i \, . \tag{F.11}$$